Sight-Reading Music Theory:
A Thought Experiment on Improving Pedagogical Efficiency

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Abstract

Nearly a thousand years ago, Guido d'Arezzo imagined "sight-reading written music," in support of which he developed new technology (including the staff and solmisation) and new pedagogy (including his use of the hymn *Ut queant laxis* and the exercise *Alme rector*) [1, pp. 459-475]. Sight-reading enabled Guido to turn novices into proficient singers in just one year, or at most two—an efficiency improvement of at least 5:1, and perhaps 10:1, compared to earlier singer-training practices [1, pp. 445-447].

As a "thought experiment," this paper imagines extending Guido's core innovations to enable "sight-reading music theory." It is proposed that developing this skill may have the potential to increase, by a factor similar to Guido's, the rate at which students become proficient music theorists. The thought experiment uses a hypothetical system for displaying and controlling musical information called JIMS Isomorphic Music System (JIMS).

Keywords: Music notation, alternative notations, alternative controllers, alternative tunings, music control interfaces, microtonality, music theory, pedagogy, parsimony.

1. Thought Experiments and Parsimony

Thought experiments [2] are an important tool in the advancement of science. In a thought experiment, one explores the impact of a presumed "law of nature" on a scenario that is stripped to its essential elements. Thought experiments are *not* intended to prove anything; rather, they provide an alternative way of thinking about a problem's context, which may lead to new experimental designs and, eventually, hard data from lab experiments.

Those laws and thought experiments are best which exhibit the most *parsimony*: the ability to explain much with little [3]. Parsimonious systems tend to provide both simplicity and power. Einstein (a noted thought experimenter) said [4] "The supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience" (usually paraphrased as "things should be made as simple as possible, but no simpler").

This paper presumes a law, presents a scenario, and invites the reader to imagine the interaction of the law and the scenario's elements.

- **Presumed law**: That students can learn music theory[1] fastest when the internal consistency of music's intervallic relationships is maximally exposed.
- **Scenario**: The reader is teaching music theory to former high-school band students who have no experience with the traditional technologies of music theory pedagogy such as the piano keyboard, grand staff, figured bass, Roman numerals, etc.; also, the reader's success as a theory teacher is measured by the extent to which her students subsequently advance the state of the musical arts.

2. Musical Invariances

In musical systems, parsimony in the display and control of musical information can be increased by exposing *invariance* (*i.e.*, consistency under transformation) through *isomorphism* (*i.e.*, consistency of mapping). Two invariant musical properties were particularly useful in the design of JIMS: *transpositional invariance* and *tuning invariance*.

It is well-known that the familiar interval patterns of music theory—such as the diatonic scale, the major triad, and the V–I cadence—retain their essential nature even when transposed to different octaves or keys. They can be said to have the property of *transpositional invariance*.

Tuning invariance—the property of being consistent under transformation among tunings—will be discussed in greater depth in Section 4.2.1 below.

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1 "Music theory" is defined herein to mean the theory of the West's tonal music in the Common Practice era, with potential expansions as described in section 4.2.1.

3. Musical Isomorphisms

To capitalize on parsimony, music's invariant properties can be exposed isomorphically (*i.e.*, with consistent mappings from property to symbol).

In the design of JIMS, four isomorphisms were particularly useful: the *isomorphic note-layout*, *tonic solfa*, the *chromatic staff*, and the *tonnetz.*

3.1 Isomorphic Pitch-Layout

A button-field is a two-dimensional arrangement of pitch-controlling buttons. A pitch-layout is a mapping of pitches to a button-field.

On a button-field with an *isomorphic pitch-layout*—*i.e.*, an *isomorphic button-field*—pressing any two buttons that have the same geometric relationship to each other sounds the same musical interval. The shape of the imaginary line connecting the centers of those two buttons is the *button-field shape* of that interval.

Any given interval's button-field shape is the same everywhere within an isomorphic button-field (edges aside). Therefore, on an isomorphic button-field, any given sequence or combination of musical intervals—a scale, a melody, a chord, a chord progression, etc.—has the same shape within a key and across all keys [5]. That is, isomorphic button-fields are transpositionally invariant.

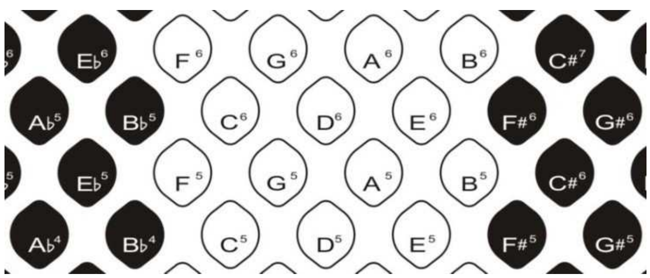

Figure 1: A button-field optimized for use with Wicki's isomorphic pitch-layout (see section 3.1.1).

3.1.1 Wicki's Pitch-Layout

For mathematical and practical reasons that are beyond the scope of this paper [6], JIMS uses the isomorphic note-layout first described by Kaspar Wicki in 1896 [7] (see Figure 1 above).

Wicki's pitch-layout has been used in squeeze-box instruments such as concertinas, bandoneons, and bayans. The use of such instruments to competently perform a wide variety of complex and challenging chromatic music shows that the Wicki pitch-layout is a practical music-control interface.

An example of the button-field's transpositional invariance is shown in Figure 2 below.

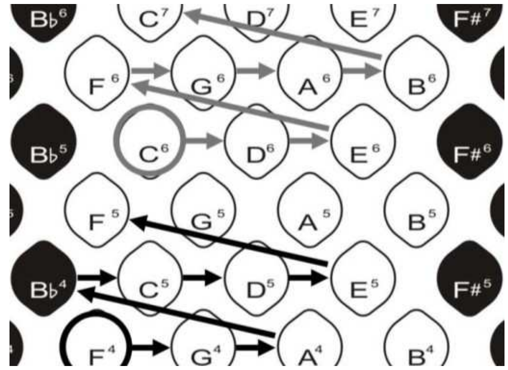

Figure 2: Isomorphism of the diatonic scale in C Major (grey lines) and F Major (black lines). Different key, same pattern.

Figure 2 shows the isomorphism of the diatonic scale in C Major (grey lines) and F Major (black lines). The scale's button-field shape is the same in both keys, and indeed in every key. The invariance of the isomorphic button-field across all keys—for all tonal structures, not just scales—offers a significant increase in parsimony vs. traditional music-control interfaces.

3.2 Tonic Solfa

The ability to express patterns of intervals with transpositional invariance is central to tonal music theory. Many systems have been invented to meet that need, including North Indian *sargam*, the Nashville Numbering System, the association of the numbers 0 to 11 with the notes of the chromatic scale, the association of mode degrees with Roman numerals, and solmization.

Of the many variants of solmization, JIMS uses *movable Do with a La-based minor* [8, 9]. The absolute pitch associated with a given tonic solfa *note*, named with a unique syllable, changes with every key, but the tonal relationships among the notes are the same in every key (*i.e.*, are invariant). Tonic solfa gives a unique name to seven diatonic notes and ten chromatic notes (five flats & five sharps), as shown in Table 1 below.

Table 1: Tonic solfa interval names.

Flat	Natural	Sharp
(De)	Do	Di
Ra	Re	Ri
Me	Mi	(My)
(Fe)	Fa	Fi
Se	So	Si
Le	La	Li
Te	Ti	(Di)

The names in parentheses are novel to JIMS. They name flat (De, Fe) and sharp (My, Di) notes that are enharmonic with diatonic notes in 12-tone equal temperament (12-TET), but not in many other tunings. The need for these "extra" notes will become clear later in this paper.

3.2.1 JIMS Note-Layout

Figure 1 and Figure 2 above label buttons with traditional pitch names. However, JIMS' parsimony can be significantly increased by associating its buttons with tonic solfa notes instead of pitches, as shown in Figure 3 below.

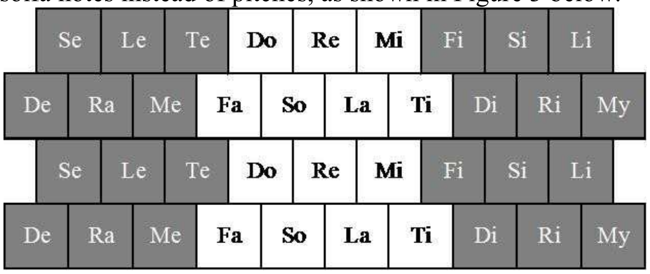

Figure 3: JIMS note-layout.

The physical arrangement of rectangular buttons shown in Figure 3 is that of a standard computer keyboard, demonstrating JIMS' suitability for computer-based instruction.

To play in C Major, one must indicate (by performing a user-interface control gesture that is beyond the scope of this paper) that Do should sound the pitch C. To indicate C minor, one must indicate that *La* should sound a C (hence *La-based minor*). To indicate B-flat Lydian, one must indicate that Fa should sound a B-flat, and so on, for other keys and modes. The same control gesture can be applied during performance to effect a key change. This real-time electronic transposition is easily accomplished by the computers to which computer keyboards are usually attached.

3.2.2 Fingering Invariance

The combination of tonic solfa, electronic transposition, and an isomorphic note-layout exposes another invariant property of music, *fingering invariance* [10], in which one presses precisely the same buttons—not just the same *pattern* of buttons—to play a given pattern of intervals in any key. For example, to play the diatonic tertian triad rooted on the tonic of the diatonic scale's Ionian mode (*i.e.*, DoMiSo), one always presses the Do, Mi, and So buttons, irrespective of the current key.

Fingering invariance increases parsimony, but it creates a problem: it is not compatible with traditional staff notation. A new staff is needed, on which note-heads represent "movable" tonic solfa notes rather than absolute pitches.

3.3 JIMS Staff

Combining the chromatic staff (first described by Roualle de Boisgelou in 1764 [11]) with tonic solfa gives JIMS staff [12] (see Figure 4, below), which is compatible with JIMS button-field.

JIMS staff maps a tonic solfa note to each line and space of the chromatic staff.

On the chromatic staff, pairs of notes that are enharmonically equivalent in 12-TET can be distinguished by using pointed note-heads that indicate the direction of deviation from the diatonic scale in Do-mode (Ionian/major).

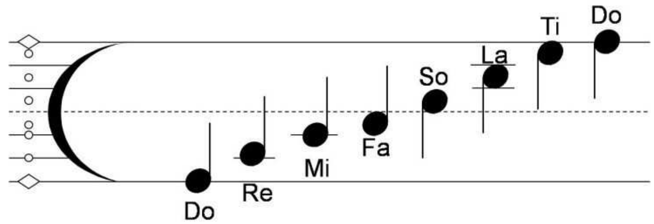

Figure 4: JIMS staff.

For example,

- Ri (an augmented second above Do), being sharpened relative to Do-mode's diatonic major second Re, would have an upward-pointing note-head, while
- Me (a minor third above Do), being flattened relative to Do-mode's diatonic major third Mi, would have a downward-pointing note-head.

Such pairs of notes are mapped to the same JIMS staff location. For example, Ri and Me are both mapped to the space between the Re-line and the Mi-line.

At the far left of JIMS staff, a stack of *scale dots* indicates the set of notes that are in the *current scale*. The diamond-shaped scale dot indicates the *current tonic* (and thereby the current mode of the current scale). In Figure 4, the scale dots indicate Do-mode of the diatonic scale. The horns of the crescent-shaped clef sign always point to Do, whatever the key, register, scale, or mode. JIMS staff is the same for every key and octave. A pitch class name (A, Bb, C#, etc.), pitch (A4), or frequency (440Hz), can be placed to the left of the tonic indicator to tie the notation to a specific key, or (preferably) left unspecified.

All other symbols from traditional music notation (*e.g.*, meter, rhythm, dynamics, etc.) are retained in JIMS notation with their traditional meanings. A detailed description of JIMS staff notation (including indicators of key changes, mode changes, and scale changes) is beyond the scope of this paper (see [12]).

JIMS approach to scales and modes works with non-diatonic scales, too. Their notes are mapped to the tonic solfa syllables such that the mapping is as symmetrical as possible around Re. For example, the double harmonic scale is mapped to Re–Me–Se–So–La–Te–Ra, of which the Re-mode is the double harmonic major, and the So-mode is the double harmonic minor (also known as Gypsy Minor).

JIMS' combination of solfeggio and a corresponding key-independent staff, *per se*, is not novel; it is exactly what Guido intended, used, and described a thousand years ago. To quote Pesce ([1], p. 23): "For Guido, *D* as a sounding pitch is not fixed, but *D* as a notational symbol has a very fixed meaning—its intervallic context." That is, broadly speaking, Guido's system notated relative pitch ("movable Ut," so to speak), not absolute pitch.[2]

JIMS staff and JIMS button-field are both isomorphic with the underlying structure of music, so they are also isomorphic with each other, as shown in Figure 5 below.

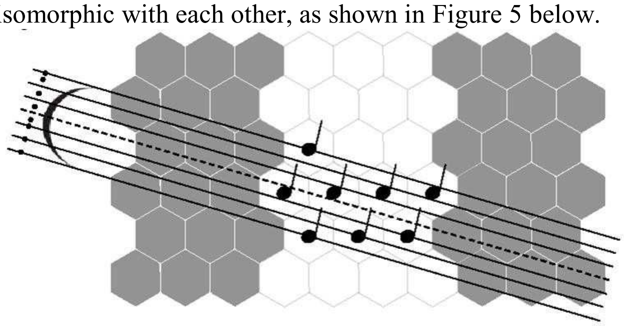

Figure 5: Isomorphism of JIMS button-field and staff. The scale dots should all be white, not black as shown.

JIMS staff and button-field are also isomorphic with another geometric depiction of music's underlying structure: the tonnetz.

3.4 Tonnetz

The tonnetz (also known as the "harmonic lattice" and "Euler-lattice") is a map of tonal space that links pitches or chords by perfect fifths and major thirds. First described by Euler in 1739, it has been used as a tool for describing tonal pitch space by music theorists ever since [13]. The tonnetz is usually depicted as a rectangular lattice, as in Figure 6 below.

F#3 — C#4 — G#4 — D#5 — A#5 — E#6
D3 — A3 — E4 — B4 — F#5 — C#6
Bb2 — F3 — C4 — G4 — D5 — A5
Gb2 — Db3 — Ab3 — Eb4 — Bb4 — F5

Figure 6: Traditional tonnetz, with perfect fifths along the horizontal axis and major thirds along the vertical axis.

However, the axes of the tonnetz can be aligned with the hexagonal axes of JIMS button-field, as shown in Figure 7 below.

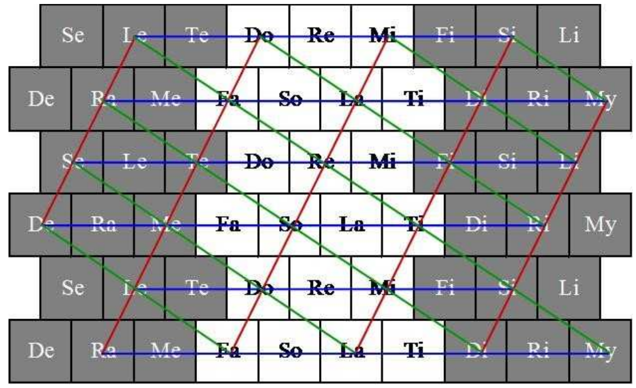

Figure 7: Tonnetz aligned with JIMS button-field.

The red lines denote axes of perfect fifths; blue lines denote axes of major thirds; green lines denote axes of minor thirds.

The triangles formed by the intersecting red, blue, and green lines above enclose the major and minor triads of tonal space. These triads are shown in Figure 8 below (major triads are capitalized; minor triads are not).

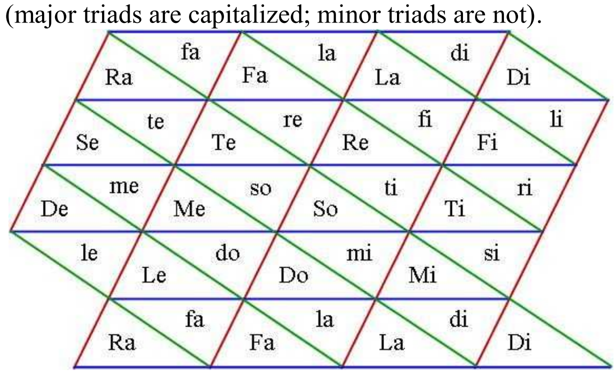

Figure 8: JIMS button-field's map of tonal space.

In his writings [1], Guido continually stressed the importance of helping students understand a piece's intervallic context. This intervallic context is, of course, the very heart of tonal music theory. JIMS' isomorphic keyboard, tonnetz, chromatic staff, scale dots, and tonic indicator expose this intervallic context to students' eyes and fingers in a geometrically-consistent manner, which is right in line with Guido's innovations and stated intent.

3.4.1 Reducing memorization-load

To distinguish among different things, different terms are needed. For example, in JIMS, a *scale* is a unique collection of tonic solfa notes, while a *mode* is a cycle around an octave of those notes. Hence, the "major scale" and "minor scale" are not scales at all, but rather are different modes of the same (diatonic) scale. The meaning of these fundamental terms must simply be memorized.

However, JIMS was designed in part to minimize the students' memorization load by using a small number of fundamental terms in as many compatible ways as possible, rather than defining new terms for specialized uses.

[2] As Guido might have said, *plus ça change, plus c'est la même chose* (except, of course, that he would have said it in Latin).

For example, JIMS:

- Names modes after their tonics (*e.g*. Do-mode, Re-mode, and Mi-mode) instead of Greek localities (*e.g.*, Ionian, Dorian, and Phrygian, respectively).
- Uses the self-explanatory phrases "tonic-bounded" and "tonic-centered" to describe the range of notes around the tonic, instead of the traditional terms "authentic" and "plagal," respectively.
- Names key-independent chords and progressions (*e.g.*, V–I) using (a) Arabic numerals when no specific mode is intended, and (b) tonic solfa names when a specific mode is intended. For example, the mode-specific diatonic chord progression Do–Fa–La–Re–So–Do (*i.e.*, I–IV–vi–ii–V–I) ends in a 5–1 cadence.
- Names chord inversions with tonic solfa-based chord names such as 3Do (indicating a diatonic tertian Do triad with its third in the bass) and 5So7 (indicating a dominant 7 chord on So with the 5th in the bass) rather than figured bass.

There are a number of advantages to the minimization of jargon. First, it requires the memorization of fewer unique terms. Second, its terms are modular, enabling compound concepts to be described using compound terms. For example, terms in the form "[scale name] Xx-mode" can describe any mode of any scale, such as "diatonic La-mode" or "double harmonic Re-mode." Third, and most importantly, it strongly links the concepts that share the same terms.

4. Simple and Powerful

Like chemistry's Periodic Table of the Elements, JIMS' parsimony gives it the potential to be both *simple* and *powerful*.

4.1 Simple

A thought experiment is not intended to prove anything, but rather establishes a conceptual framework within which lab experiments can be designed and executed. Only live human trials can determine whether "sight-reading music theory," using a JIMS-like technology (*i.e.*, a technology that maximizes the parsimonious exposure of musical invariance through isomorphism) can deliver efficiency improvements approaching those attained by Guido. Because JIMS is only a thought experiment, no educational materials exist for it, so no such trials are currently possible.

Instead, readers are invited to imagine teaching an introductory music theory class using JIMS and to ask themselves what might be gained and lost thereby.

One could imagine, for example, teaching students about secondary dominants using JIMS. When a piece's scale dots indicate that it is in diatonic Do-mode, and a Re7 chord is denoted, there's no ambiguity: Re7 is the dominant of the dominant, signaling a temporary tonicization of So. If the tonicization were not temporary, JIMS notation would indicate (by means that are beyond the scope of this paper; see [12, pp. 31]) a change of key "up a perfect fifth" (thereby changing the pitches associated with the tonic solfa notes), and the "Re7" chord would have been notated as So7 in that new key. Either way, the tonal meaning of the denoted chord ought to be entirely clear.

Likewise, every occurrence of the diatonic Do–Fa–So–Do chord progression will look exactly like every other (inversions aside), and its root movement on JIMS button-field will follow the tonnetz in exactly the same pattern (octave transpositions aside), intimately relating movement on a button-field with movement in tonal space.

The authors submit that, compared to the traditional technology of music theory pedagogy, JIMS-like technologies:

- present music theory students with fewer things (*i.e.*, facts, rules, symbols, and gestures) to learn;
- inter-relate these things into consistent cognitive chunks; and
- expose the logical consistency of the relationships among these chunks to students' senses of sight and touch in addition to hearing.

4.2 Power

Just as JIMS' *simplicity* arises from its parsimony, so does its *power*. JIMS offers two kinds of power: *creative power* and *expressive power*.

4.2.1 Creative Power: Dynamic Tonality

JIMS' creative power arises from its isomorphic exposure of *tuning invariance*, which is the property of being consistent under transformation among tunings. First documented in 2007 [*ibid*., 6], the details of tuning invariance are beyond the scope of this paper.

In brief, isomorphic note-layouts are tuning invariant, such that any given sequence or combination of musical intervals has the same fingering in every tuning of a given temperament, such as the *syntonic temperament* [*ibid.*, 6] on which JIMS is based, *irrespective of the number of notes into which the tuning subdivides the octave*. For example, the pentatonic scale's Do-mode (major) is always played by pressing the same buttons— "Do Re Mi So La" —whether the current tuning of the syntonic temperament has 5, 7, 12, 17, 19, or even 31 notes per octave, and whether the tuning is equal (*e.g.*, 12-TET), unequal (*e.g.*, ¼-comma meantone), or irregular (*e.g.*, a well-temperament or a rank-2 *p*-limit Just Intonation).

Tuning invariance enables performers to retaining consistent fingering while changing tuning smoothly, in real time, along (for example) the *syntonic tuning continuum* [*ibid.*, 6], on which are found many tunings used in the history of Western culture [15, 16], and by the indigenous music of some non-Western cultures [17, 18, 19, 20, 21]. Consistent fingering makes it easy to explore these alternative tunings and the relationships among them.

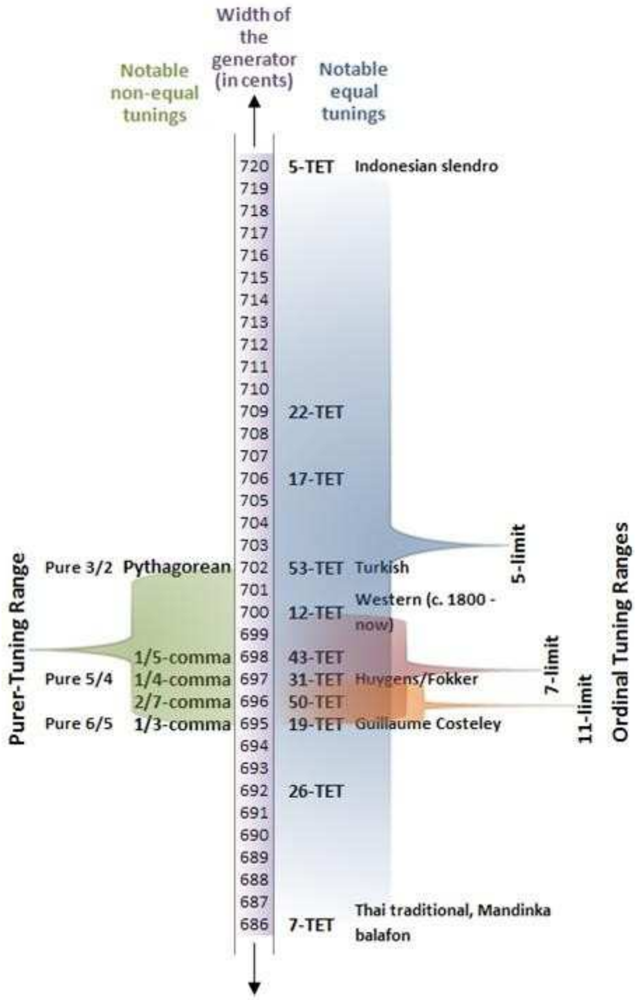

Figure 9: The syntonic tuning continuum.

In Figure 9 above, "N-TET" stands for "N-Tone Equal Temperament." 12-TET (or simply "equal temperament") is a syntonic tuning in which P5 = 700 cents.

Continuous real-time tuning changes, called *polyphonic tuning bends*, are the basis of *tuning progressions* and *temperament modulations*. When performing such a tuning bend, the current timbre's partials can be tempered to maintain constant alignment with the current tuning's notes [22], thereby providing the option of consonance in any tempered tuning and also enabling dynamic timbre effects such as *sonance* (dynamic changes to the consonance/dissonance of any simultaneous combination of notes) and *primeness* (changing the percentage of a timbre's energy that is invested in partials that share a given prime factor). Changing tuning and timbre together enables *dynamic tonality* [23], which can be seen as a generalization of the relationship between the Harmonic Series and Just Intonation to include the tuning continua of many temperaments, including the syntonic temperament on which JIMS is based.

By expanding the framework of tonal harmony to embrace these new structural resources, dynamic tonality offers art musicians—and music theorists—new creative power, to which JIMS provides an interface.

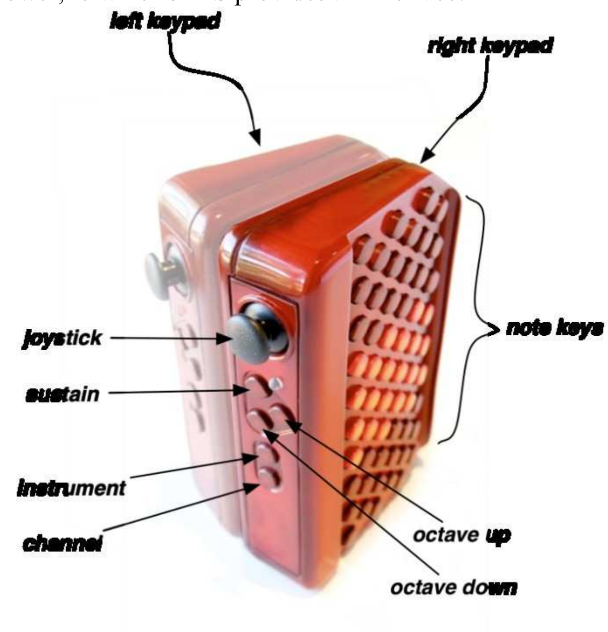

Figure 10: A prototype Thummer.

4.2.2 Expressive Power: JIMS Controllers

To be musically interesting, physical embodiments of JIMS note-layout—*i.e.*, "JIMS controllers"—must have the potential to meet or exceed the expressive power of traditional musical instruments [24]. One such JIMS controller is the Thummer [25], of which a prototype is shown in Figure 10 above.

On the Thummer, three full octaves, of 19 buttons per octave, fit within the span of a single hand's fingers. Each fingertip can press either a single note-controlling button or a combination of adjacent buttons sounding, among them, a P4, a P5, a sus2, or a sus4. This enables the performance of complex chords with just the four fingers of a single hand.

Combining two such button-fields—one for each hand—produces a tiny controller, as shown in Figure 10.

The Thummer's tiny size allows it to be supported by a brace affixed to one forearm, as shown in Figure 11 below, with the other arm remaining completely free. The use of a brace frees both hands' fingers to play their respective button-fields, and both hands' thumbs to control tiny joysticks like those on a video game controller. The instrument can also contain internal motion sensors, like those in Nintendo's Wii Remote.

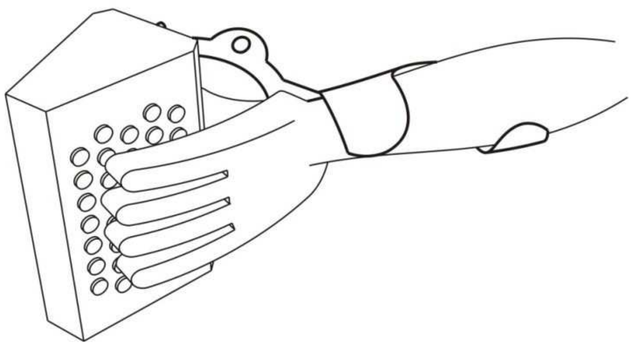

Figure 11: Sketch of prototype Thummer forearm brace.

Each of the Thummer's two joysticks has two axes, and a six-axis internal motion sensor has six more (obviously), so altogether the Thummer can offer up to ten degrees of freedom. Each such variable can be devoted to the control of any expressive parameter, such as pitch bend, vibrato speed, vibrato depth, brightness, reverberation, etc. Using the Thummer's two three-octave button-fields, a musician could perform an *expressive* solo with one hand while playing accompaniment with the other, all on the same tiny instrument.

No other polyphonic instrument of which the authors are aware offers similar *expressive potential* (*i.e.*, the control over as many degrees of freedom).

Another potential JIMS controller is the standard alphanumeric computer keyboard. What it lacks in expressiveness, it makes up for in ubiquity, being even more commonplace than the piano keyboard or guitar.

Apple's iPhone[3] could also act as a JIMS controller. Its multi-touch screen is big enough only for the buttons controlling two octaves of the diatonic scale (perhaps plus Te and Fi), which should be enough to be interesting. Its built-in motion sensors could control expressive variables such as pitch bend and volume, so that shaking the iPhone in one direction would produce vibrato, and in another, tremolo.

5. Metrics

Before Guido invented sight-singing, how was a singer's "quality" (*i.e.*, value to a church's choir) measured? One contributory metric, among many, might have been "the number of songs he could sing from memory." Neither Guido's methods nor his students would have scored well on this metric. Sure, they could *sing*, but they *didn't know any songs!* If a choir could not provide such a singer with expensive, hand-copied-on-parchment music, he could add little or no value. Yet the value of sight-reading trumped these reasonable concerns.

This paper's thought experiment similarly challenges the metrics by which the "quality" of a music theorist might be measured. On the one hand, the use of a JIMS-like technology could reduce the value of some currently-valuable skills by making them irrelevant or trivially easy. Arguably, examples include the ability to transpose notation among clefs and keys; to perform a functional analysis of a notated work; to identify of key centers and key relationships; to recognize modulation to closely related keys; and so on.

On the other hand, some skills that are not highly valued today could become more highly valued when using a JIMS-like technology. One such example might be the ability to recognize the new structural resources of dynamic tonality, including tuning effects (such as *polyphonic tuning bends*, *tuning progressions*, and *temperament modulations*) and timbre effects (such as *primeness* and *sonance*) [22]. These italicised terms are as unfamiliar to music educators today as "sight-singing," "written music," and "solmization" were in Guido's day. This is hardly surprising, because dynamic tonality's exploration requires elements of JIMS, and at present JIMS is just a thought experiment—just as Guido's "sight-singing" once was.

6. Previous Work

Holland [26] found that significant speed-of-learning was improved by the use of an isomorphic note-layout for display but not control, but used too small a sample size (five people) for the results to be conclusive. Bergstrom [27], also using an isomorphic note-layout for display but not control, stated that in a trial of roughly 30 people, "most viewers absorb[ed] the basics quickly."

To the authors' knowledge, no large and rigorous trial of any JIMS-like system has yet been performed.

7. Thought Experiments

As has been emphasized throughout this paper, JIMS is currently just a thought experiment. No JIMS-compatible course materials, notation software, instruments, examinations, or other pedagogical paraphernalia exist for it. Therefore, clearly, this paper cannot and does not propose that JIMS be used today in music theory pedagogy.

Rather, JIMS is presented here to support these four arguments:

1. That, just as Guido's sight-reading of *notes* improved the efficiency of singer-training by at least 5:1, so also might the sight-reading of *theory* improve the efficiency of theorist-training by a similar factor.
2. That such sight-reading might require new technology and pedagogy…just as was true in Guido's time.
3. That the technology needed to enable the sight-reading of theory can be assembled from off-the-shelf ideas (*e.g.*, tonic solfa, the tonnetz, the chromatic staff, and the isomorphic keyboard).
4. That the sight-reading of theory might have "unexpected" consequences—such as enabling an

[3] www.apple.com/iphone

expansion of tonality, facilitating the development of exceptionally-expressive musical instruments, and driving a re-examination of the metrics used to measure the skills of music theory—that can accelerate advancement of the state of the art.

Readers are invited to conduct their own thought experiments by asking related questions of themselves and their peers. Such questions might include:

1. What content (*i.e.*, knowledge and skill) in tonal music theory is independent of its traditional technology (including instruments and notation)?
2. What different purposes does music theory's technology serve (*e.g.*, performance, composition, communication, analysis, pedagogy)?
3. What criteria [c_1, c_2, …, c_n] are relevant to the comparison and contrasting of alternative musical technologies, and how should those criteria be weighted [w_1, w_2, …, w_n] in a criteria-consolidating metric [w_1c_1 + w_2c_2 +…+ w_nc_n] for each said purpose?
4. Would the existence of hypothetical electronic translation tools, which could translate music-theoretical ideas from one technology to another, alter said metric for any said purpose?
5. Would said metric depend on whether the relevant students were music theory majors, music majors not concentrating on theory, non-major music students, or non-academic musical hobbyists?
6. What experiments should be used to objectively measure the merits of alternative technologies for each said purpose using said metric?

8. Future Research

Future research could address the development of a royalty-free, open-source reference design for JIMS controllers; the implementation of JIMS controllers based on the computer keyboard, Apple iPhone, or other existing hardware; the design and implementation of JIMS notation software and translators; the development of JIMS-based music theory pedagogy and course materials; human trials that measure any difference in music theory students' learning speed and/or outcomes when using the traditional or JIMS technologies; and/or answers to any of the thought-experiment questions listed above.

9. Contributions

This paper contributes:

- Sections 1-6: A presentation of JIMS Isomorphic Music System (JIMS) as a "thought experiment" on how *maximizing the parsimonious and isomorphic exposure of invariance in the display and control of musical information* might enable the sight-reading of music theory, thereby increasing the efficiency of music theory pedagogy.
- Section 7: A suite of "thought experiment" questions that could focus subsequent discussion of alternative technologies.
- Section 8: A proposal for a collaborative research program that would measure any differences in efficiency and outcome between JIMS and traditional technologies.

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